Probability Theory

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Prerequisites for Probability

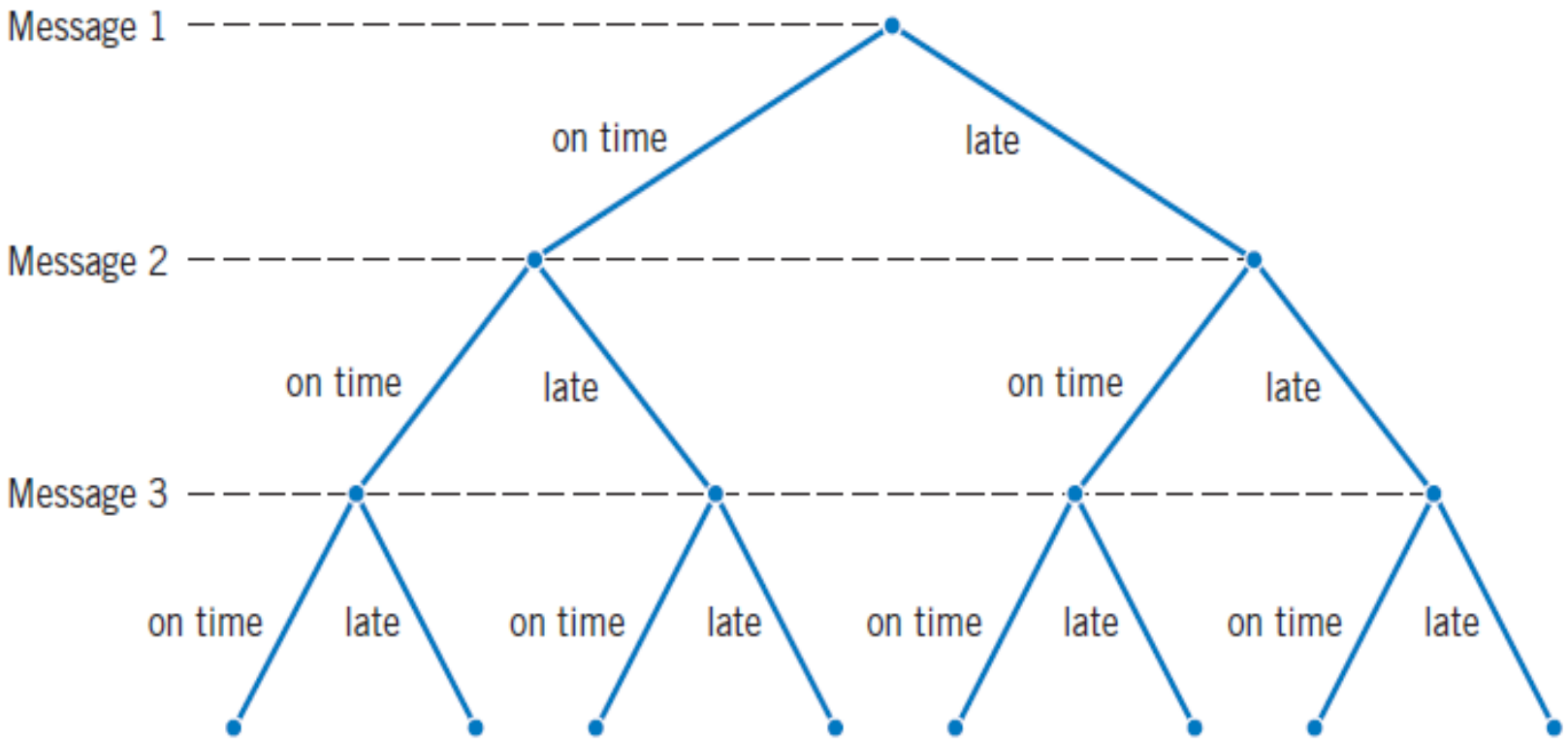
- **Counting Principle**
- **Permutations**
- **Combinations**

Counting

- **Multiplication Principle:** Suppose an event **E** can occur in ***m*** different ways and associated with each way of occurring of **E**, another event **F** can occur in ***n*** different ways, then the total number of occurrence of the two events in the given order is $m \times n$.

Example : Each message in a digital communication system is classified as to whether it is received within the time specified by the system design. If three messages are classified, use a tree diagram to represent the sample space of possible outcomes.
Each message can be received either on time or late.

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Counting

- **Addition principle:** If an event **E** can occur in ***m*** ways and another event **F** can occur in ***n*** ways, and suppose that both can not occur together, then **E** or **F** can occur in ***m*** **+** ***n*** ways.

Ex: Padmavathi wants to go to Vijayawada from Hyd. She can choose from 3 bus services or 2 train services to head from Hyd to Vijayawada. How many ways are there for Padma to go to Vijayawada?

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From Hyd to Vijayawada=3+2=5 ways.

- **Padmavathi wants to go to Vizag from Hyd. She can choose from 3 bus services or 2 train services to head from Hyd to Vijayawada. From there, she can choose from 4 bus services or 3 train services to Vijayawada to Vizag. How many ways are there for Padma to go to Vizag?**



- **Padmavathi wants to go to Vizag from Hyd. He can choose from 3 bus services or 2 train services to head from Hyd to Vijayawada. From there, she can choose from 4 bus services or 3 train services to Vijayawada to Vizag. How many ways are there for Padma to go to Vizag?**

From Hyd to Vijayawada=3+2=5 ways

Vijayawada to Vizag=4+3=7 ways

Padma to go to Vizag = 5*7=35 ways



Permutations

Permutation of n different objects :The number of permutations of ***n*** objects taken all at a time, denoted by the symbol

$${}^{n}P_{n} \quad \textbf{or} \quad P_n^n = n! = n \times (n-1) \times (n-2) \times \ldots\ldots 2 \times 1.$$

Permutation of r elements selected from n different objects :

The number of permutations of subsets of r elements selected from a set of n different elements is

$$P_r^n = n \times (n-1) \times (n-2) \times \cdots \times (n-r+1) = \frac{n!}{(n-r)!}$$

${}^{n}P_{r}$

No. of arrangement $=$

$$n.(n-1)(n-2)....(n-r+1) = \frac{?!}{?!}$$

$$= \frac{n.(n-1)(n-2)....(n-r+1)(n-r)....2.1}{(n-r)(n-r-1)...2.1}$$

$$= \frac{n!}{(n-r)!} = {}^{n}P_{r} \qquad (r \leq n)$$

Combinations

The number of combinations, subsets of size r that can be selected from a set of n elements, is denoted as $\binom{n}{r}$ or C_r^n and

$$C_r^n = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Example: There are 12 boys and 14 girls in math class. Find the number of ways Dr. Padmavathi can select a team of 3 students from the class to work on a group project. The team is to consist of 1 girl and 2 boys.

Order, or position, is not important.

Example: There are 12 boys and 14 girls in math class. Find the number of ways Dr. Padmavathi can select a team of 3 students from the class to work on a group project. The team is to consist of 1 girl and 2 boys.

Order, or position, is not important. Using the multiplication counting principle.

$$\underset{\textit{Boys}}{12_{C_2}} \times \underset{\textit{Girls}}{14_{C_1}} = \frac{12 \times 11}{2 \times 1} \times \frac{14}{1} = 924$$

Why We Learn Probability?

- Nothing in life is certain.
- In everything we do, we gauge (measure) the chances of successful outcomes, from business to medicine to the weather.
- It provides a bridge between descriptive and inferential statistics.

- What is a random Experiment?

- What is a random Experiment?

An experiment is called **random experiment** if, when (experiment) repeated under the same conditions, it is such that the outcome cannot be predicted with certainty but all possible outcomes can be determined prior to the performance of the experiment.

Sample Space

- The set or collection of all possible outcomes or events of a random experiment is called the **Sample Space** denoted by **S.**
- **Elements of the sample space are called sample points.**
- **http://www.regentsprep.org/regents/math/algebra/APR4/Lsample.htm**

Sample Space

Discrete Sample Space : A discrete sample space is one with a finite or countably infinite number of possible values.

i. Inspecting a lightbulb. {Defective, Nondefective}.
ii. Rolling a die,S={1,2,3,4,5,6}
iii. Asking for opinion about a new car model. {Like, Dislike, Undecided}.
iv. The number of bytes counted past a point in a network in a second is a discrete sample space with the possible outcomes between 0 and the bandwidth of the Link in bytes per second.

An automobile manufacturer provides vehicles equipped with selected options. Each vehicle is ordered
With or without an automatic transmission
With or without air conditioning
With one of three choices of a stereo system
With one of four exterior colors.

If the sample space consists of the set of all possible vehicle types, what is the number of outcomes in the sample space?

An automobile manufacturer provides vehicles equipped with selected options. Each vehicle is ordered
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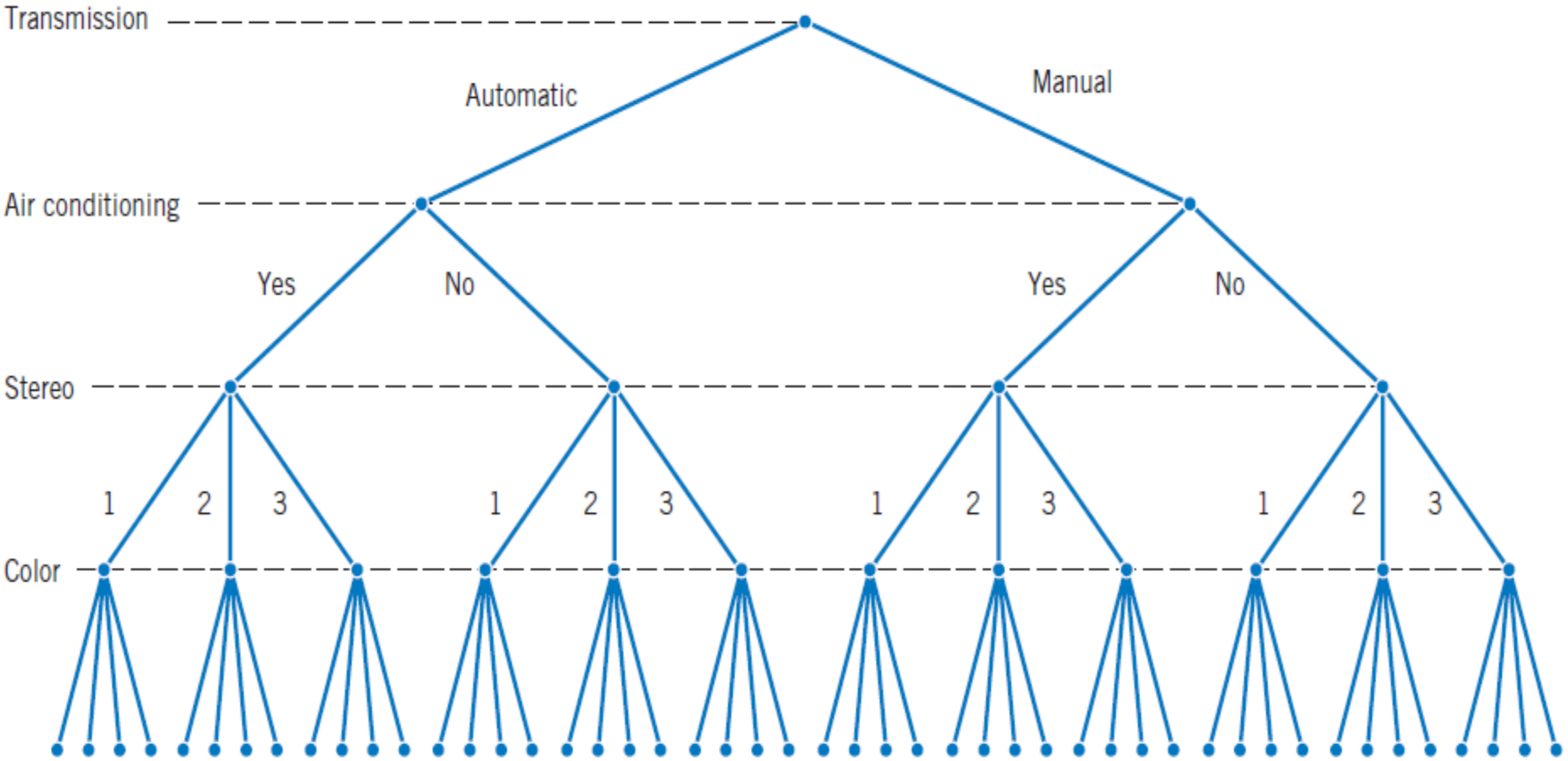


Figure 2-6 Tree diagram for different types of vehicles with 48 outcomes in the sample space.

When we say the events are Mutually exclusive?

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Events are said to be mutually exclusive, if the happening of any one of the events in a trial excludes the happening of any one of the others.

Mutually exclusive Events:

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Ex:

From a well shuffled pack of cards, we pick up one card at random and would like to know whether it is a king or queen.

The selected card will be either a king or queen. But it cannot be both king or queen.



Equally Likely

- A set of events are said to be equally likely if no one of them expected to occur in preference to others in any single trial of the random experiment.

Ex: When card is drawn from a pack, any card may be obtained . In this trial all the 52 elementary events are equally likely.

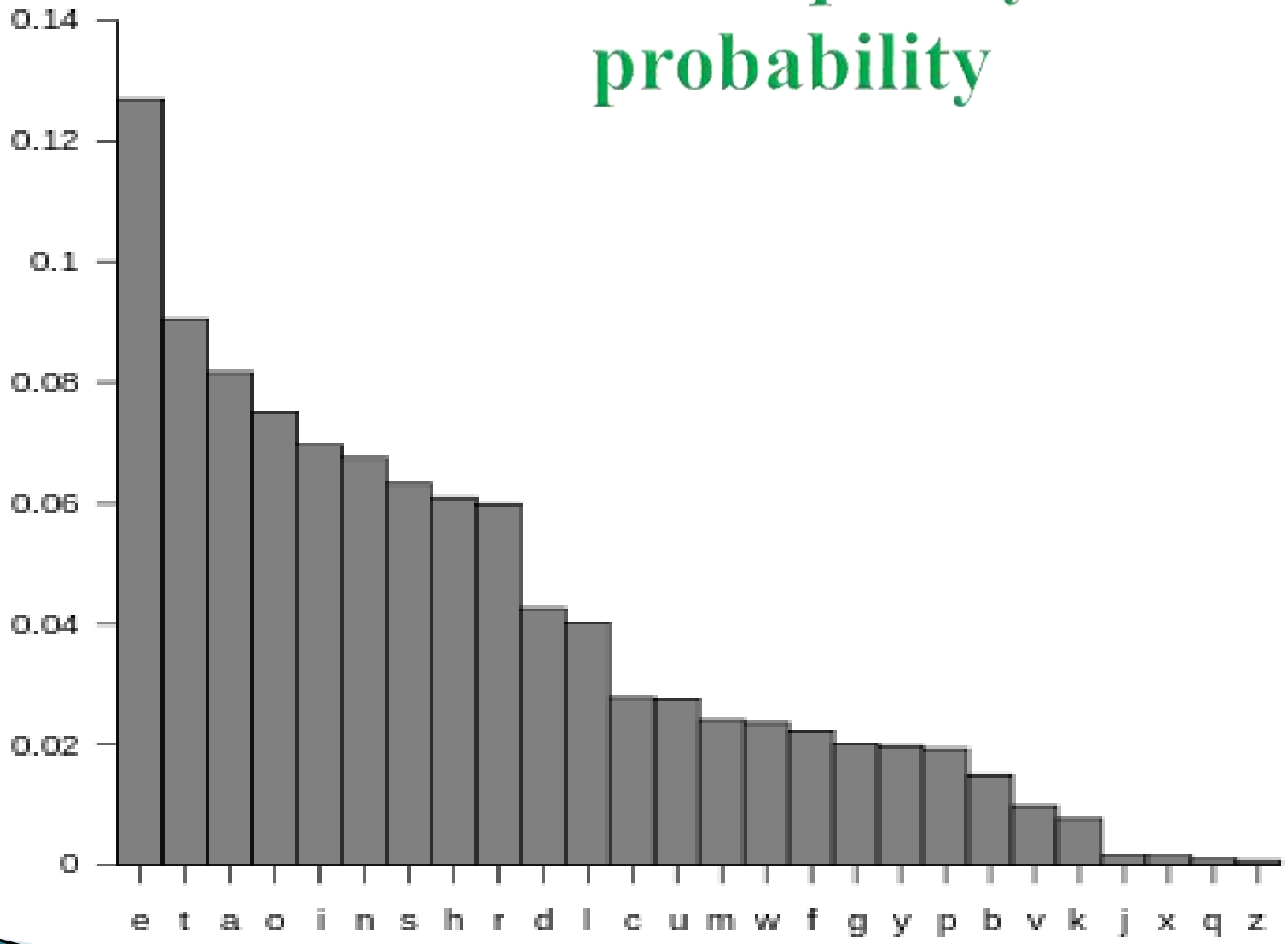
By letter

Letter	Frequency
a	0.08167
b	0.01492
c	0.02782
d	0.04253
e	0.12702
f	0.02228
g	0.02015
h	0.06094
i	0.06966
j	0.00153
k	0.00772
l	0.04025
m	0.02406
n	0.06749
o	0.07507
p	0.01929
q	0.00095
r	0.05987
s	0.06327
t	0.09056
u	0.02758
v	0.00978
w	0.02360
x	0.00150
y	0.01974
z	0.00074

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Letter frequency or probability



When we say events are Independent ?

Independent Events

- Two events are said to be independent if the occurrence or non occurrence of one event has no influence on the occurrence or non occurrence of the other event .

Independent Events

- Two events are said to be independent if the occurrence or non occurrence of one event has no influence on the occurrence or non occurrence of the other event .

Ex:

1. **If we draw a card from a pack of well shuffled cards and replace it before drawing second card, then the second draw is independent of the first draw.**

2. If the first draw is not replaced then the second draw is dependent on the first draw.



- **Favourable Events:** The events, which are favourable to a particular event of an experiment are called favourable events.

Ex: When a die is rolled, getting 2,4 and 6 are favourable events to the event "getting an even number".

- **Success and Failures :**Events which are favourable to particular event of an experiment are called success and the remaining are called failures with respect to that event.

Ex: When a die is rolled, getting 2,4 and 6 are success for the event "getting an even number". And getting 1,3 and 5 are failures.

Probability

If the sample S of an experiment consists of finitely many outcomes that are eqally likely, then the probability P(A) of an event A is

$$P(A)=\frac{\text{Number of points in A}}{\text{Number of points in S}}=\frac{\text{Number of favourable cases to A}}{\text{All possible cases to S}}$$

$$=\frac{n(A)}{n(S)}.$$

Axioms of Probability

(i) Axiom of non-negetivity : $0 \leq P(E) \leq 1$ for every subset E of S.

(ii) Axiom of certainity : P(S)=1.

(iii) Axiom of union : If E_1, E_2 are disjoint subset of S, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2).$$

$\rightarrow$ If S is infinite (has infinitely many points),

if $E_1, E_2, ..., E_n, ...$ are mutually exclusive (disjoint) subset of S, then

$$P(E_1 \cup E_2 \cup ..... \cup E_n) = P(E_1) + P(E_2) + ... + P(E_n)$$

$$P(E_1 \cup E_2 \cup .....) = P(E_1) + P(E_2) + ...$$

Some Elementary Theorems of Probability

Complementary rule

Th-1: For an event A and its complement A^c or A' or $\bar{A}$ in a sample space S,

$$P(A^c) = 1 - P(A).$$

Proof:

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Proof: Since A and A^c are mutually exclusive subsets of S,

we have $S = A \cup A^c$ and $A \cap A^c = \phi$.

From Axiom 2 and 3 we have,

Proof: Since A and A^c are mutually exclusive subsets of S,

we have S= A $\cup$ A^c and A $\cap$ $A^c = \phi$.

From Axiom 2 and 3 we have,

$$1=P(S)=P(A \cup A^c)$$

$$= P(A) + P(A^c).$$

$$\therefore \quad P(A^c) = 1 - P(A).$$

$\rightarrow$ Five coins are tossed simultaneously. Find the probability of the event A: At least one head turns up. Assume that the coins are fair.

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So $n(S)=2^5 = 32.$

$$P(A^c) = P(\text{no heads turn up})= \frac{1}{32}.$$

$$\text{Thus} \quad P(A) = 1 - P(A^c) = 1 - \frac{1}{32} = \frac{31}{32}.$$

Addition Rule for Mutually Exclusive Events

For mutually exclusive events $A_1, A_2, ..., A_n$ in a sample space S,

$$P(A_1 \cup A_2 \cup ..... \cup A_n) = P(A_1) + P(A_2) + ... + P(A_n).$$

Addition Rule for Mutually Exclusive Events

Th-2: For mutually exclusive events $A_1, A_2, ..., A_n$ in a sample space S,

$$P(A_1 \cup A_2 \cup ..... \cup A_n) = P(A_1) + P(A_2) + ... + P(A_n).$$

Proof: Let $A = A_1 \cup A_2 \cup ..... \cup A_n$,

we prove this by induction on 'n' .

For n=2, and from axiom 3 we have

$$P(A_1 \cup A_2) = P(A_1) + P(A_2).$$

Assume for n-1

i.e, $$P(A_1 \cup A_2 \cup ..... \cup A_{n-1}) = P(A_1) + P(A_2) + ... + P(A_{n-1}).$$

$$P(A_1 \cup A_2 \cup ..... \cup A_{n-1} \cup A_n) = P(A_1 \cup A_2 \cup ..... \cup A_{n-1}) + P(A_n)$$

$$= P(A_1) + P(A_2) + ... + P(A_{n-1}) + P(A_n).$$

- If the probability that on any workday a garage will get 10–20, 21–30, 31–40, over 40 cars to service is 0.28, 0.35, 0.25, 0.12, respectively, what is the probability that on a given workday the garage gets at least 21 cars to service?

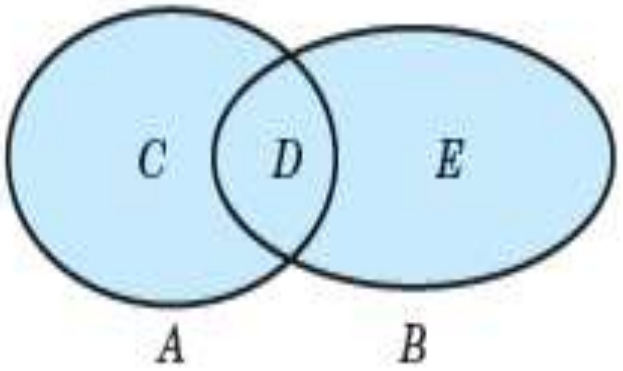
Ans: 0.35 + 0.25 +0.12 = 0.72.

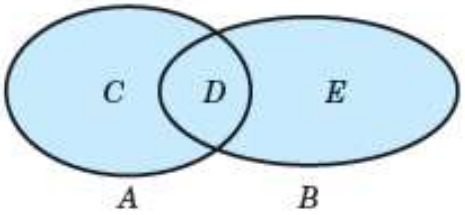
In many cases, events will not be mutually exclusive. Then we have **Addition Rule for Arbitrary Events.**

Addition Rule for Arbitrary Events

Th -2:For events A and B in a sample space S,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$





Let $C=A \cap B'$, $D=A \cap B$, $E=A' \cap B$.

So C,D,E make up $A \cup B$ (i.e, $A \cup B = C \cup D \cup E$)

and are mutually exclusive (disjoint).

Hence by, *Addition Rule for Mutually Exclusive Events*

$$P(A \cup B) = P(C \cup D \cup E) = P(C) + P(D) + P(E).$$

Since $P(A) = P(C \cup D)$

$$= P(C) + P(D)$$

And $P(E) = P(B) - P(D)$

$$= P(B) - P(A \cap B),$$

Hence $P(A \cup B) = P(C) + P(D) + P(E)$

$$= P(A) + P(B) - P(A \cap B).$$

In tossing a fair die, what is the probability of getting an odd number or a number less than 4?

Let *A be the event "Odd number"={1,3,5} and*
B the event "Number less than 4."={1,2,3}

$$P(A \cup B) = P(A)+P(B) - P(A \cap B)$$

$$= \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{2}{3}$$

Conditional Probability. Independent Events

- Often it is required to find the probability of an event B under the condition that an event A occurs.

- This probability is called the **conditional probability of B given A and is denoted** by B|A.

In this case A serves as a new (reduced) sample space, and that probability is the fraction of which corresponds to $A \cap B$.

Thus

$$P(B|A)=\frac{P(A\cap B)}{P(A)} \qquad [P(A)\neq 0].$$

Similarly, the conditional probability of A given B is

$$P(A|B)=\frac{P(A\cap B)}{P(B)} \qquad [P(B)\neq 0].$$

Example: Toss a fair coin 3 times.

(a) What is the probability of 3 heads?

(b) Suppose we are told that the first toss was heads. Given this information how should we compute the probability of 3 heads?

Example: Toss a fair coin 3 times.

(a) What is the probability of 3 heads?

Sol: Sample space S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}. All outcomes are equally likely, so P(3 heads) = 1/8.

(b) Suppose we are told that the first toss was heads. Given this information how should we compute the probability of 3 heads?

Sol: We have a new sample space: A = {HHH, HHT, HTH, HTT}. All outcomes are equally likely, so P(3 heads given that the first toss is heads) = 1/4.

Ex of conditional probability:

Suppose that somebody secretly rolls two fair six-sided dice, and we must predict the outcome.

Let A be the value rolled on die 1

Let B be the value rolled on die 2

What is the probability that $A = 2$?

Suppose it is revealed that $A+B \leq 5$. *Find* $P(A=2 \mid A+B \leq 5)$.

Ex of conditional probability:

Table 1						
+	B=1	2	3	4	5	6
A=1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

What is the probability that $A = 2$?

Table 1 shows the sample space. $A = 2$ in 6 of the 36 outcomes, thus $P(A=2) = {}^{6}\!/_{36} = {}^{1}\!/_{6}$.

Suppose it is revealed that $A+B \le 5$. Table 2 shows that $A+B \le 5$ for 10 outcomes. For 3 of these, $A = 2$. So the probability that $A = 2$ *given that* $A+B \le 5$ is $P(A=2 \mid A+B \le 5) = {}^{3}\!/_{10} = 0.3$.

Table 2						
+	B=1	2	3	4	5	6
A=1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
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Multiplication Rule

Th - 4: If A and B are events in a sample space S and

Thus $P(A \cap B) = P(A)\ \mathrm{P(B|A)} = P(B)\mathrm{P(A|B)}$.

Ex:In producing screws, let *A mean "screw too slim" and B "screw too short." Let P(A)=0.1 and let the conditional* probability that a slim screw is also too short be P(B|A)= 0.2.

What is the probability that a screw that we pick randomly from the lot produced will be both too slim and too short?

- In producing screws, let *A mean "screw too slim" and B "screw too short." Let P(A)=0.1 and let the conditional* probability that a slim screw is also too short be P(B|A)= 0.2.

- What is the probability that a screw that we pick randomly from the lot produced will be both too slim and too short?

$$P(A \cap B) = P(A)\ \mathrm{P(B|A)} = (0.1)(0.2) = 0.02 = 2\ \%.$$

Independent of the Event

Def: If A and B are events in a sample space S.

Then A and B are said to be Mutually independent if

$$P(B|A) = P(B)$$

and $P(A|B) = P(A)$ where $P(A) \neq 0, P(B) \neq 0$.

Th : (Multiplicative Rule for independent events)

If A and B are events in a sample space S and

Thus $P(A \cap B) = P(A)\ P(B)$.

Ex: A bag contains 8 red and 6 blue balls. Two drawing of each 2 balls are made. Find the probability that first drawing gives two red balls and second drawing gives 2 blue balls , if the balls drawn first are replaced before the second draw.

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Sol:

A: event of drawing 2 red balls B: event of drawing 2 blue balls

$$P(A) = \frac{8C_2}{14C_2} \quad \text{and} \quad P(B|A)=P(B) = \frac{6C_2}{14C_2}$$

$$P(A \cap B)=P(A)\,P(B)= \frac{60}{1183}$$

Total Probability

When we talk about total probability we are talking about a partition space.

A partition of any sample space is nothing but simply a collection of events that are disjoint say *B*1,*B*2…,*Bn* and the union of all these events is equal to the sample space.

$$A = A \cap S = A \cap (B_1 \cup B_2 \cup .... \cup B_n)$$

$$\therefore A = (A \cap B_1) U (A \cap B_2) \cup ..... \cup (A \cap B_n)$$

$$Then \ P(A) = P(A \cap B_1) + P(A \cap B_2) + ... + P(A \cap B_n)$$

$$= P(B_1)P(A \mid B_1) + P(B_2)P(A \mid B_2) + ... + P(B_n)P(A \mid B_n)$$

At a certain university, 4% of men are over 6 feet tall and 1% of women are over 6 feet tall. The total student population is divided in the ratio 3:2 in favour of women. What is the probability of student over 6 feet.

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P(W)=3/5
P(M)=2/5
T=Student over 6 feet
P(T/W)=1/100
P(T/M)=4/100
P(T)=P(W)P(T/W)+P(M)P(T/M)=11/500

Example 1: We draw two cards from a deck of shuffled cards without replacement. Find the probability of getting the second card a queen.

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- A: the event of getting the first card a queen

  $P(A)=4/52$

- B:the event that the first card is not a queen

  $P(B)=1-P(A)=48/52$

- E:the event that the second card is a queen
- $P(E/A)=3/51$ and $P(E/B)=4/51$

- $P(E) = P(A)P(E/A)+P(B)P(E/B)$

  $=4/52* 3/51+48/52*4/51$

  $=1/13$

Bayes Theorem

Let $E_1, E_2, ..., E_n$ *are n* mutually exclusive events such that $P(E_i) > 0 \; (i = 1, 2, .., n)$ in a sample space S and A is any other event of S intersecting with every E_i such that P(A)>0.

If $P(E_1), P(E_2), ..., P(E_n)$ and $P(A|E_1), ..., P(A|E_n)$ are known then

$$P(E_k \mid A) = \frac{P(E_k)P(A|E_k)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + ... + P(E_n)P(A|E_n)}$$

Companies B1, B2, B3 produce 30%, 45% and 25% of the cars respectively. If it is known that 2%, 3% and 2% of the cars produced from B1, B2 and B3 are defective. If a car purchased is found to be defective. What is the probability that their car is produced by company B1, B2 and B3?

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	P(Bi),i=1,2,3	P(D/Bi)	P(Bi) P(D/Bi)	
B1	0.3	0.02	0.006	0.2449
B2	0.45	0.03	0.0135	0.55102
B3	0.25	0.02	0.005	0.20408
			0.0245	1

Ex of Baye`s theorem (verification):

Suppose that somebody secretly rolls two fair six-sided dice, and we must predict the outcome.

Let *A* be the value rolled on die 1

Let *B* be the value rolled on die 2

Table 1						
+	B=1	2	3	4	5	6
A=1	2	3	4	5	6	7
2	3	4	5	6	7	8
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4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Define X= {2, 3,..., 12}, obtained by considering the sum of two dice.

Suppose Y is a random variable which takes on the value D if the two dice are the same (i.e, we throw doubles) and N, otherwise.

$$\textit{Then verify} \quad p(\mathrm{D}\,|\,4) = \frac{p(4\,|\,D)\,p(\mathrm{D})}{p(4)}.$$

Suppose Y is a random variable which takes on the value D if the two dice are the same (i.e, we throw doubles) and N, otherwise.

$$p(D) = 6/36 = 1/6$$

$$p(N) = 30/36 = 5/6$$

$$p(4) = 3/36 = 1/12 \qquad p(4 \mid D) = 1/6$$

$$\frac{p(4 \mid D)p(\mathrm{D})}{p(4)} = \frac{1/6 \cdot 1/6}{1/12} = 1/3 = p(\mathrm{D} \mid 4)$$

$$p(D \mid 4) = 1/3$$

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Table						
+	B=1	2	3	4	5	6
A=1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

In a TV Game show, a contestant selects one of three doors; behind one of the doors there is a prize, and behind the other two there are no prizes. After the contestant selects a door, the game-show host opens one of the remaining doors, and reveals that there is no prize behind it. The host then asks the contestant whether they want to SWITCH their choice to the other unopened door, or STICK to their original choice.

Is it probabilistically advantageous for the contestant to SWITCH doors, or is the probability of winning the prize the same whether they STICK or SWITCH? (Assume that the host selects a door to open, from those available, with equal probability).

Answer: Without loss of generality, let events A, B, C correspond to the prize being behind the selected, opened, and remaining door respectively, and let H_B denote the event that the host opens door B. Want to compare $\mathrm{P}(A|H_B)$ (STICK) with $\mathrm{P}(C|H_B)$ (SWITCH). Now $\mathrm{P}(A) = \mathrm{P}(B) = \mathrm{P}(C) = 1/3$, and we are given that $\mathrm{P}(H_B|A) = 1/2$, $\mathrm{P}(H_B|B) = 0$ and $\mathrm{P}(H_B|C) = 1$. Then the general version of Bayes' theorem gives

$$\mathrm{P}(A|H_B) = \frac{\mathrm{P}(H_B|A)\mathrm{P}(A)}{\mathrm{P}(H_B)} = \frac{\mathrm{P}(H_B|A)\mathrm{P}(A)}{\mathrm{P}(H_B|A)\mathrm{P}(A) + \mathrm{P}(H_B|B)\mathrm{P}(B) + \mathrm{P}(H_B|C)\mathrm{P}(C)} = \frac{\frac{1}{2}\frac{1}{3}}{\frac{1}{2}\frac{1}{3} + 0\frac{1}{3} + 1\frac{1}{3}} = \frac{1}{3}$$

so $\mathrm{P}(A|H_B) = 1/3$, and similarly $\mathrm{P}(C|H_B) = 2/3$. so it is advantageous to SWITCH.

- A diagnostic test has a probability 0.95 of giving a positive result when applied to a person suffering from a certain disease, and a probability0.10 of giving a (false) positive when applied to a non-sufferer. It is estimated that 0.5 % of the population are sufferers. Suppose that the test is now administered to a person about whom we have no relevant information relating to the disease (apart from the fact that he/she comes from this population). Calculate the following probabilities:

(a) that the test result will be positive;

(b) that, given a positive result, the person is a sufferer;

(c) that, given a negative result, the person is a non-sufferer;

(d) that the person will be misclassified.

Answer: Let $T \equiv$ "Test positive", $S \equiv$ "Sufferer", $M \equiv$ "Misclassified." Then $\mathrm{P}(T|S) = 0.95$, $\mathrm{P}(T|S') = 0.10$, $\mathrm{P}(S) = 0.005$. Hence

$$\text{(a)}\quad \mathrm{P}(T) = \mathrm{P}(T|S)\mathrm{P}(S) + \mathrm{P}(T|S')\mathrm{P}(S') = (0.95 \times 0.005) + (0.1 \times 0.995) = 0.10425.$$

$$\text{(b)}\quad \mathrm{P}(S|T) = \frac{\mathrm{P}(T|S)\mathrm{P}(S)}{\mathrm{P}(T|S)\mathrm{P}(S) + \mathrm{P}(T|S')\mathrm{P}(S')} = \frac{0.95 \times 0.005}{(0.95 \times 0.005) + (0.1 \times 0.995)} = 0.0455.$$

$$\text{(c)}\quad \mathrm{P}(S'|T') = \frac{\mathrm{P}(T'|S')\mathrm{P}(S')}{\mathrm{P}(T')} = \frac{0.9 \times 0.995}{1 - 0.10425} = 0.9997.$$

$$\text{(d)}\quad \mathrm{P}(M) = \mathrm{P}(T \cap S') + \mathrm{P}(T' \cap S) = \mathrm{P}(T|S')\mathrm{P}(S') + \mathrm{P}(T'|S)\mathrm{P}(S) = 0.09975.$$

A new phone, X1 has been announced and it is what you've been waiting for, all along. You decide to read the reviews before buying it. From past experiences, you've figured out that good reviews mean that the product is good 90% of the time and bad reviews mean that it is bad 70% of the time. Upon glancing through the reviews section, you find out that the X1 has been reviewed 1269 times and only 127 of them were bad reviews. What is the probability that, if you order the X1, it is a bad phone?

For the solution, let's use the following abbreviations.

- BP - Bad Phone
- GP - Good Phone
- GR - Good Review
- BR - Bad Review

From the given data, $Pr(BP|BR) = 0.7$ and $Pr(GP|GR) = 0.9$. Using this, Pr(BP—GR) = 1 - Pr(GP—GR) = 0.1.
Hence,

$$\begin{aligned} Pr(BP) &= Pr(BP|BR) \cdot Pr(BR) + Pr(BP|GR) \cdot Pr(GR) \\ &= 0.7 \cdot \frac{127}{1269} + 0.1 \cdot \frac{1269 - 127}{1269} \\ &= 0.160 \end{aligned}$$

Probability provides a bridge between descriptive and inferential statistics.

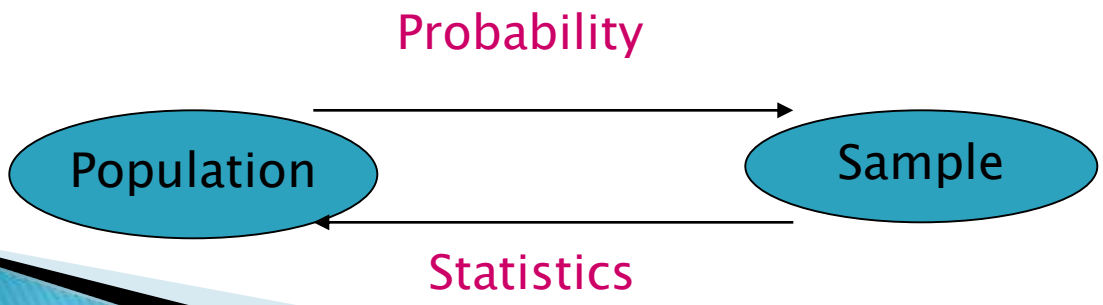
Descriptive Statistics	Inferential Statistics
Descriptive statistics includes statistical procedures that we use to describe the population we are studying.	Inferential statistics studies a statistical sample, and from this analysis is able to say something about the population from which the sample came.
The data could be collected from either a sample or a population, but the results help us organize and describe data. That is, the results cannot be generalized to any larger group	A Chi-square or T-test, for example, can tell us the probability that the results of our analysis on the sample are representative of the population that the sample represents.
Frequency distributions, measures of central tendency (mean, median, and mode), and graphs like pie charts and bar charts that describe the data are all examples of descriptive statistics.	**Examples of inferential statistics include linear regression analyses, logistic regression analyses, ANOVA, correlation analyses, structural equation modeling, and survival analysis, to name a few.**

“probability is common sense reduced to calculation”

-Laplace

- Suppose I know exactly the proportions of car makes in California. Then I can find the probability that the first car I see in the street is a Ford. This is **probabilistic reasoning** as I know the population and predict the sample

- Now suppose that I do not know the proportions of car makes in California, but would like to estimate them. I observe a random sample of cars in the street and then I have an estimate of the proportions of the population. This is **statistical reasoning**



Big O

- Big O notation is used in Computer Science to describe the performance or complexity of an algorithm.

- Big O specifically describes the worst-case scenario, and can be used to describe the execution time required or the space used (e.g. in memory or on disk) by an algorithm.

O(1)

- O(1) describes an algorithm that will always execute in the same time (or space) regardless of the size of the input data set.

- **O(1)** means that, no matter how much data, it will execute in constant time.

O(N)

O(N) describes an algorithm whose performance will grow linearly and in direct proportion to the size of the input data set.

- *Big O notation will always assume the upper limit where the algorithm will perform the maximum number of iterations.*

$O(N^2)$

$O(N^2)$ represents an algorithm whose performance is directly proportional to the square of the size of the input data set.

This is common with algorithms that involve nested iterations over the data set. Deeper nested iterations will result in $O(N^3)$, $O(N^4)$ etc.

$O(2^N)$

- $O(2^N)$ denotes an algorithm whose growth doubles with each additon to the input data set.
- The growth curve of an $O(2^N)$ function is exponential - starting off very shallow, then rising meteorically.

O(logN)

- O(logN) - if the target value is lower than the value of the probe element it will perform the operation against the lower half. It will continue to halve the data set with each iteration until the value has been found or until it can no longer split the data set.
- An input data set containing 10 items takes one second to complete, a data set containing 100 items takes two seconds, and a data set containing 1000 items will take three seconds. Doubling the size of the input data set has little effect on its growth as after a single iteration of the algorithm the data set will be halved and therefore on a par with an input data set half the size.

Constant Multiplication:

If $f(n) = c.g(n)$, then $O(f(n)) = O(g(n))$; where c is a nonzero constant.

Polynomial Function:

If $f(n) = a_0 + a_1.n + a_2.n^2 + \text{—-} + a_m.n^m$, then $O(f(n)) = O(n^m)$.

Summation Function:

If $f(n) = f_1(n) + f_2(n) + \text{—-} + f_m(n)$ and $f_i(n) \leq f_{i+1}(n)\ \forall\ i=1, 2, \text{—-}, m$, then $O(f(n)) = O(max(f_1(n), f_2(n), \text{—-}, f_m(n)))$.

Logarithmic Function:

If $f(n) = \log_a n$ and $g(n)=\log_b n$, then $O(f(n))=O(g(n))$;

all log functions grow in the same manner in terms of Big-O.

Mathematical Examples

- If n = 10,
- log(10) = 1;
- 10 = 10;
- 10log(10)=10;
- 10^2=100;
- 2^{10}=1024;
- 10!=3628800;

If n=20,

log(20) = 2.996;

20 = 20;

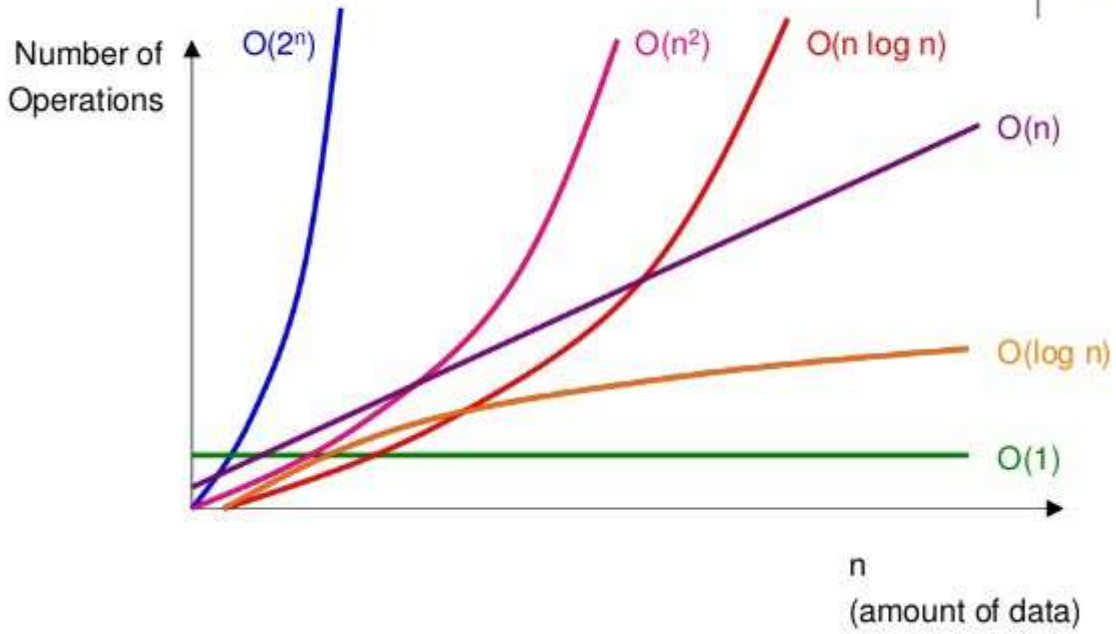
20log(20)=59.9;

20^2=400;

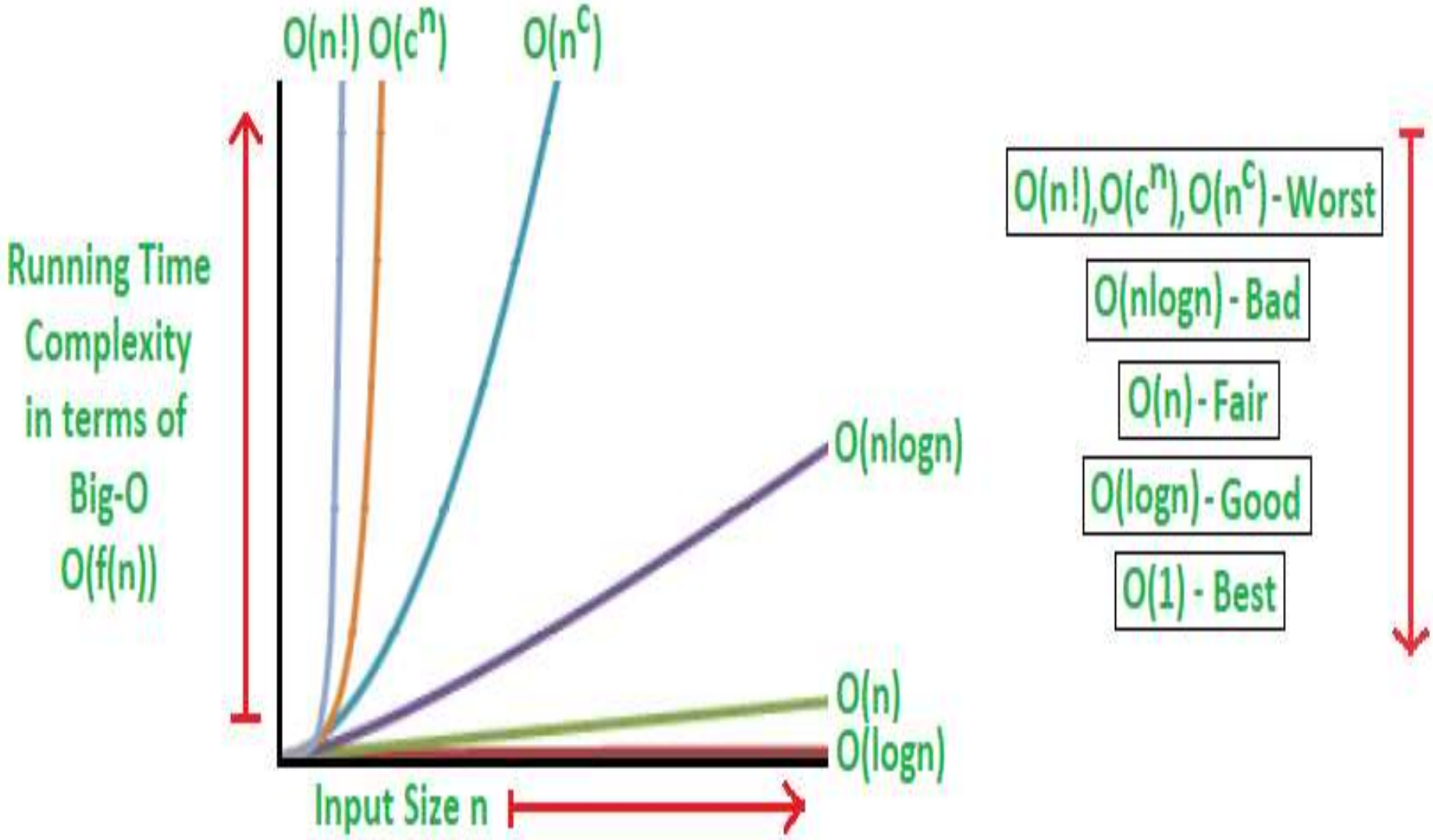
2^{20}=1048576;

20!=2.432902e+18^{18};

Comparing Big O Functions



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O(n!) O(c^n)
O(n^c)
Running Time Complexity in terms of Big-O O(f(n))
O(nlogn)
O(n)
O(logn)
Input Size n
O(n!),O(c^n),O(n^c) - Worst
O(nlogn) - Bad
O(n) - Fair
O(logn) - Good
O(1) - Best

Common Data Structure Operations

Data Structure	Time Complexity								Space Complexity
	Average				Worst				Worst
	Access	Search	Insertion	Deletion	Access	Search	Insertion	Deletion	
Array	Θ(1)	Θ(n)	Θ(n)	Θ(n)	O(1)	O(n)	O(n)	O(n)	O(n)
Stack	Θ(n)	Θ(n)	Θ(1)	Θ(1)	O(n)	O(n)	O(1)	O(1)	O(n)
Queue	Θ(n)	Θ(n)	Θ(1)	Θ(1)	O(n)	O(n)	O(1)	O(1)	O(n)
Singly-Linked List	Θ(n)	Θ(n)	Θ(1)	Θ(1)	O(n)	O(n)	O(1)	O(1)	O(n)
Doubly-Linked List	Θ(n)	Θ(n)	Θ(1)	Θ(1)	O(n)	O(n)	O(1)	O(1)	O(n)
Skip List	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(n)	O(n)	O(n)	O(n)	O(n log(n))
Hash Table	N/A	Θ(1)	Θ(1)	Θ(1)	N/A	O(n)	O(n)	O(n)	O(n)
Binary Search Tree	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(n)	O(n)	O(n)	O(n)	O(n)
Cartesian Tree	N/A	Θ(log(n))	Θ(log(n))	Θ(log(n))	N/A	O(n)	O(n)	O(n)	O(n)
B-Tree	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(log(n))	O(log(n))	O(log(n))	O(log(n))	O(n)
Red-Black Tree	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(log(n))	O(log(n))	O(log(n))	O(log(n))	O(n)
Splay Tree	N/A	Θ(log(n))	Θ(log(n))	Θ(log(n))	N/A	O(log(n))	O(log(n))	O(log(n))	O(n)
AVL Tree	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(log(n))	O(log(n))	O(log(n))	O(log(n))	O(n)
KD Tree	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(n)	O(n)	O(n)	O(n)	O(n)